CHRIS OGHUMAH BALARABE

SOFTWARE ENGINEERING GRADUATE | MSc CLOUD COMPUTING STUDENT

Ireland | 089 949 5107 | oghumahchris@gmail.com
github.com/Balarabe18 | linkedin.com/in/chris-oghumah-138685292/

PROFILE

MSc Cloud Computing student at National College of Ireland with a First Class Honours Higher Diploma in Computing (Software Engineering). Academic and personal project experience with Java, Spring Boot, REST APIs, Python, SQL and machine learning, supported by practical AWS, Linux, Docker and Kubernetes work. Interested in graduate software engineering, cloud, application support and data-platform opportunities in Ireland.

TECHNICAL SKILLS

Programming: Java, Python, C, JavaScript, SQL; object-oriented programming

Backend & data: Spring Boot, REST APIs, MySQL, Django, NumPy; data preparation and machine-learning evaluation

Cloud & containers: AWS EC2, S3, Lambda, Elastic Beanstalk; Linux, Amazon Linux, Docker, Kubernetes (kind)

Testing & tools: Git, GitHub, Postman, pytest, pytest-cov, AWS CLI, IntelliJ IDEA

Distributed computing: Open MPI, POSIX threads, performance benchmarking, strong and weak scaling

EDUCATION

MSc Cloud Computing | National College of Ireland

September 2026 - Present | Expected completion: August 2027

Current modules: Cloud Architectures, Cloud Platform Programming, AI in the Cloud, GreenOps and Sustainability.

Higher Diploma in Science in Computing (Software Engineering)

National College of Ireland | January 2024 - April 2025
Result: First Class Honours

SELECTED ACADEMIC & PERSONAL PROJECTS

Barbershop Management System

Java | Spring Boot | MySQL | REST APIs | Thymeleaf | JavaScript

Built a web application for Customer, Barber and Administrator roles, supporting appointment booking, rescheduling, cancellation and confirmation.

Developed backend services and REST API endpoints with MySQL persistence. Tested API behaviour using Postman.

Adult Income Machine-Learning Project

Python | Machine learning | AI in the Cloud

Prepared an income-classification workflow and compared logistic regression, decision tree and random forest models.

Explored hyperparameter tuning and evaluated accuracy, precision, recall, F1 and ROC-AUC. Examined performance across sex and race groups to assess fairness and communicate limitations.

SELECTED PROJECTS - CONTINUED

AWS Serverless & Application Deployment

Python | Lambda | S3 | EC2 | Elastic Beanstalk | Django | Linux

Created and invoked Python Lambda functions, validated inputs and used execution logs to troubleshoot behaviour.

Configured an S3 event trigger with an incoming-file prefix and tested matching and non-matching object paths. Deployed a Django application to Elastic Beanstalk and worked with EC2, Linux command-line tools and Python virtual environments.

Container & Kubernetes Performance Labs

Docker | Kubernetes | kind | C | Linux

Containerised a C benchmark and compared CPU and memory performance under different resource constraints.

Deployed replicated Nginx workloads to a local Kubernetes cluster and observed automatic Pod replacement after deleting a Pod. Configured CPU and memory requests and limits, then inspected logs and deployment details to analyse resource behaviour.

Parallel Computing & Scaling Benchmarks

C | POSIX threads | Open MPI | Performance analysis

Built and ran parallel benchmarks to measure execution time, speedup and efficiency across worker counts.

Compared compute and memory workloads under strong and weak scaling to analyse hardware and resource bottlenecks. Practised MPI ring messaging, ping-pong, distributed mean calculation and broadcast.

Graduate Job Application Tracker - In Progress

Python | Django | HTML | CSS

Developing a Django web application to organise graduate job applications for the GreenOps and Sustainability CA.

Configured and ran the application locally, resolved development issues and passed Django system checks. Planned next stages: cloud deployment, CI/CD, automated testing, security checks and resource-efficiency evaluation. These stages are planned work.

EMPLOYMENT

SFS Associate | Sports Direct, Ireland | October 2024 - Present

Support store fulfilment and stock operations, processing customer and online-order requirements accurately to operational deadlines. Collaborate with colleagues and adapt to changing priorities while balancing part-time employment with postgraduate study.

LEADERSHIP & COMMUNICATION

Class Representative | MSc Cloud Computing, NCI

Represent MSc Cloud Computing students and communicate class concerns and feedback.

Students' Union Representative | NCI

Organised a group connecting international class representatives across NCI programmes to share student concerns and ideas for discussion with the Students' Union.